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Roll No

EE/EX-7005(3)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

FACTS

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) All part of each question are to be attempted at one place.

1. a) State the names, control/getting requirement, draw symbols, I-V characteristics of three types of switching devices used in FACTS. 7
b) State the expression for power transfers and reactive power requirement with diagram to increase the power transfer capability of transmission line using shunt compensator and state its location. 7
2. a) What are the objectives of static Var compensator in power system? Explain. 6
b) Briefly describe the way by which the transient stability is enhanced due to static Var compensator. 8
3. a) Explain the operation of GTO thyristor controlled series capacitor with line diagram and current and voltage waveforms. 8
b) What is SSR? Describe how SSR is mitigated with GCSC or TCSC? 6

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4. a) Explain the operation of SSSC and give the comparison between variable series compensation and SSSC. 7
b) Describe the operation of a unified power flow controller connected at the sending end. 7
5. a) Why the term Flexible used in FACTS? 6
b) Explain the control co-ordination of multiple controllers using linear control technique. 8
6. The voltage regulator of a SVC is designed for an ESCR of 2.0. If the slope of the control characteristics is 0.05pu, determine the transfer function of the regulator. If the operating value of ESCR is 5.0, what is the response time (to reach 95% of the final value) and the steady state change in the SVC voltage if Vref is increased by 0.05pu. Assume the system frequency to be 50Hz. 14
7. a) What are factors which limits the loading capability? 6
b) Discuss the capabilities of Shunt compensation in improving the transient stability. 8
8. Describe the following. 4+3+4+3=14
 - a) Need for FACT controller
 - b) SSR mitigation
 - c) SVC-SVC interaction
 - d) Quantitative treatment of control co-ordination

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